9th Class • Punjab Board • PCTB New Syllabus

Chapter 9: Plant Physiology

Solved notes in book order — MCQs, short questions, long questions, inquisitive questions and key points.

Chapter 9 • Units 9.1 – 9.8 • Exercise solved in book order

What this chapter is about

• Plant physiology explains how plants obtain mineral nutrients, absorb and transport water and salts, translocate food and exchange gases.
• It also explains transpiration, opening and closing of stomata, removal of wastes and osmotic adjustments in plants of different habitats.
• Important processes include osmosis, active transport, transpirational pull and the pressure flow mechanism of food transport.
• All exercise answers follow the book order. Every difference question is presented as a two-column table for easy revision.

A. Multiple Choice Questions

Select the correct option. The reason after each answer recalls the related plant process.

1. Which of the following plant nutrients is required in large amount?

a) Iron
b) Zinc
c) Potassium
d) Boron

Answer: c) Potassium — Potassium is a macronutrient and also regulates stomatal opening and closing; iron, zinc and boron are micronutrients.

2. Which element is required by plants for the formation of chlorophyll?

a) Phosphorus
b) Calcium
c) Magnesium
d) Sulphur

Answer: c) Magnesium — Magnesium is part of the chlorophyll molecule and also activates many plant enzymes.

3. The primary function of root hairs is:

a) Transport of nutrients
b) Storage of food
c) Increase surface area for absorption
d) Synthesis of proteins

Answer: c) Increase surface area for absorption — Root hairs are extensions of epidermal cells that greatly increase contact with soil water and salts.

4. Root hairs absorb salts from soil by;

a) Diffusion
b) Osmosis
c) Active transport
d) Filtration

Answer: c) Active transport — Root hairs use energy to take in more mineral ions against their concentration gradient; water enters by osmosis.

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5. Water moves from the soil into root cells by:

a) Osmosis
b) Active transport
c) Diffusion
d) Bulk flow

Answer: a) Osmosis — Water crosses the partially permeable membrane from soil water with lower solute concentration to root-cell sap with higher solute concentration.

6. The transpiration is regulated by;

a) Mesophyll
b) Guard cells
c) Xylem
d) Phloem

Answer: b) Guard cells — Changes in the turgor of guard cells open and close stomata, thereby controlling water-vapour loss.

7. Under which condition, there will be high rate of transpiration?

a) High humidity
b) Low light intensity
c) Wind
d) Waterlogged soil

Answer: c) Wind — Wind removes moist air around the leaf and maintains a strong diffusion gradient for water vapour.

8. Which ion plays a role in the opening of stomata?

a) Sodium (Na^{+})
b) Potassium (K^{+})
c) Calcium (Ca^{2+})
d) Magnesium (Mg^{2+})

Answer: b) Potassium (K^{+}) — Guard cells actively take in potassium ions; water follows by osmosis and the turgid guard cells open the stoma.

9. In most plants the food is transported in the form of:

a) Glucose
b) Sucrose
c) Starch
d) Maltose

Answer: b) Sucrose — Glucose made in photosynthesis is commonly converted to sucrose for transport through the phloem.

10. What is TRUE according to the pressure flow mechanism of food transport?

a) Water enters the source, creating pressure
b) Water is pulled from the sink
c) Movement of food in phloem is due to gravity
d) Solutes move from low to high concentration

Answer: a) Water enters the source, creating pressure — Active sucrose loading raises solute concentration at the source, so water enters by osmosis and creates high pressure.

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11. Succulent organs are present in:

a) Xerophytes b) Hydrophytes
c) Mesophytes d) Halophytes

Answer: a) Xerophytes — Many xerophytes, such as cacti, store water in soft, juicy stems or roots called succulent organs.

B. Write Short Answers

1. Define mineral nutrition in plants.

Mineral nutrition is the absorption of essential mineral elements from the soil and their use in plant growth and metabolism. These minerals are needed to make proteins, enzymes, chlorophyll, nucleic acids and other substances, and to regulate processes such as stomatal movement.

2. Define macronutrients and micronutrients and give examples.

Macronutrients are mineral elements required in large amounts, e.g. carbon, hydrogen, oxygen, nitrogen, phosphorus, potassium, sulphur, calcium and magnesium. **Micronutrients** are required in very small amounts but are still essential, e.g. iron, manganese, zinc, copper, boron, molybdenum, chlorine and nickel.

3. State the roles of nitrogen and magnesium in plants.

Nitrogen is part of amino acids, proteins, enzymes, nucleic acids and chlorophyll. It promotes leaf, seed and fruit formation; its deficiency causes poor growth and yellowing (chlorosis). **Magnesium** is part of chlorophyll, activates many enzymes and helps in fruit formation and seed germination. Its deficiency also causes chlorosis.

4. Define transpiration and its types.

Transpiration is the loss of water **in the form of vapours** from the plant surface. Its three types are **stomatal transpiration** through stomata, **cuticular transpiration** through the leaf cuticle, and **lenticular transpiration** through lenticels of woody stems. Most water loss occurs through stomata.

5. How is the transpirational pull important in plants?

Transpirational pull is the upward pulling force produced when water evaporates from leaf surfaces. It pulls the continuous xylem water column from roots to leaves, so water and dissolved mineral salts reach all aerial parts. It also helps cool the plant and maintains the water supply needed for turgor and photosynthesis.

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6. Transpiration is the loss of water from plants. Is it a harmful phenomenon? If no, what is its importance?

Transpiration is not normally harmful; it is useful because it produces the pull for upward water and mineral transport and cools the plant. It also supports photosynthesis by maintaining water supply. However, excessive loss without enough absorption can cause wilting, so it is sometimes called a **necessary evil**.

7. Differentiate between:

i. Xylem and phloem

Xylem	Phloem
Transports water and dissolved mineral salts.	Transports prepared food, mainly sucrose.
Main direction is from roots to stem and leaves.	Carries food from a source to a sink; direction depends on supply and demand.
Vessels and tracheids are dead and lignified at maturity.	Sieve tubes are living and are supported by companion cells.
Transport is driven mainly by transpirational pull.	Transport occurs by the pressure flow mechanism.

ii. Transpiration and guttation

Transpiration	Guttation
Loss of water as vapours.	Loss of water as liquid droplets.
Occurs through stomata, cuticle and lenticels.	Occurs through hydathodes at leaf tips or edges.
Common during the day when stomata are open.	Usually occurs at night or early morning when transpiration is low.
Caused by evaporation and transpirational pull.	Usually caused by root pressure forcing out xylem sap.

iii. Hydrophytes and halophytes

Hydrophytes	Halophytes
Live in freshwater or wet soil.	Live in salty water or salty marshes.
Their main problem is removal of excess water.	Their main problem is preventing water loss to the hypertonic environment and removing excess salt.

Hydrophytes	Halophytes
Often have broad floating leaves with many stomata on the upper surface.	Accumulate salts to raise cell-sap concentration and may excrete salt through leaf glands.
Example: water lily.	Examples: many sea grasses.

iv. Hydrophytes and xerophytes

Hydrophytes	Xerophytes
Live where water is plentiful.	Live in extremely dry habitats such as deserts.
Can absorb water through the whole body surface.	Have deep roots to absorb water from almost dry soil.
Usually have a thin cuticle and many exposed stomata to remove extra water.	Have a thick waxy cuticle and very few stomata to reduce water loss.
Example: water lily.	Example: cactus; many have succulent water-storing organs.

v. Lenticular transpiration and stomatal transpiration

Lenticular transpiration	Stomatal transpiration
Water vapour is lost through lenticels.	Water vapour is lost through stomata.
Occurs in the bark of woody stems.	Occurs mainly in leaves and also in young green stems.
Accounts for a very small amount of water loss.	Accounts for most of the water loss from a plant.
Lenticels remain open and are not controlled by guard cells.	Guard cells regulate it by opening and closing the stomata.

8. How do the plants of rubber and keekar excrete their wastes?

The **rubber plant** removes metabolic waste in the form of **latex**, which comes out through special pores in its bark. **Keekar (Acacia)** excretes waste in the form of **gum** through special pores. These substances are stored or removed so they do not harm active plant cells.

C. Write Answers in Detail

1. Describe the events involved in the opening and closing of stomata.

Each stoma is surrounded by two sausage-shaped guard cells. Their inner walls are thicker than their other walls. A change in their turgor changes their shape and opens or closes the pore.

Opening during daytime: Guard cells actively take in potassium ions (K^+) from surrounding epidermal cells. This raises the solute concentration of guard cells, so water enters them by osmosis. They become turgid and bean-shaped; their thick inner walls move apart and the stoma opens. Glucose made by photosynthesis also helps maintain their high solute concentration.

Closing during evening: Glucose concentration falls and potassium ions move back to epidermal cells. Water leaves the guard cells by osmosis, so they lose turgor and become flaccid. Their inner sides come together and the stoma closes. This reduces water loss when photosynthesis is not taking place.

2. Explain the internal structure of root and describe the uptake of salt and water by root.

Epidermis and root hairs: The epidermis is the outer single layer. Many of its cells form hair-like extensions among soil particles. Root hairs provide a large surface area and directly contact soil water.

Cortex: This broad region lies just inside the epidermis. Water and dissolved salts move from the epidermis through the cortex. **Endodermis** is the innermost boundary of the cortex and controls entry into the central tissues. **Pericycle** is a narrow layer just inside the endodermis. **Xylem and phloem** occupy the centre; xylem receives water and carries it to the stem.

Uptake of salts: Soil water usually has a lower salt concentration than root-hair sap. Root hairs therefore absorb additional mineral ions by **active transport**, using energy to move them against the concentration gradient.

Uptake of water: Salt absorption raises the solute concentration inside root hairs. Water then enters from the soil by **osmosis**, a passive process. Water with dissolved salts moves from cell to cell through the epidermis and cortex. It can travel through the cytoplasm of connected cells or through cell walls and intercellular spaces.

Finally, water crosses the endodermis and pericycle and enters the root xylem. The xylem of the root joins that of the stem, allowing upward transport to the leaves.

3. Describe temperature, wind and humidity as the factors affecting the rate of transpiration.

Temperature: A rise in temperature increases transpiration because water molecules gain kinetic energy and evaporate more quickly from mesophyll cell walls. Warm air can also hold more water vapour. Very high temperature, however, may close stomata to prevent excessive loss and thus reduce transpiration.

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Wind: Wind blows away the humid layer of air surrounding a leaf. This keeps the outside air drier than the air spaces inside the leaf, so water vapour diffuses out faster and transpiration increases. Still air allows moist air to collect and slows the loss.

Humidity: Humidity is the amount of water vapour in the atmosphere. High humidity reduces the difference in water-vapour concentration between the leaf and outside air, so transpiration slows. Low humidity increases this difference and speeds up transpiration.

4. Describe the mechanism of transport of water and salt in plants.

Root hairs absorb mineral salts mainly by active transport. Their accumulation makes root-cell sap more concentrated, so water enters by osmosis. The water and dissolved salts then pass through the cortex and enter the xylem.

Roots cannot push all this water to the leaves. Instead, evaporation of water from leaf surfaces produces a **transpirational pull**. Water first evaporates from moist mesophyll cell walls into leaf air spaces and then diffuses out through stomata. More water enters the mesophyll cells from the leaf xylem.

Xylem vessels contain a continuous water column from roots to leaves. **Cohesion** holds water molecules together, **adhesion** attracts them to xylem walls, and the narrow vessels help maintain the column. When water is removed from the leaf xylem, tension passes down the whole column.

As a result, the entire column is pulled upward and more water enters from the roots. Dissolved mineral salts move with this water to the stem and leaves. This is the cohesion–tension explanation of water transport.

5. Explain the mechanism of food translocation by Pressure Flow Mechanism.

Food is transported through the **phloem**, usually as sucrose, from a **source** (a place of production or supply, such as a photosynthetic leaf) to a **sink** (a place of use or storage, such as a growing tip or root). The pressure flow mechanism explains this movement:

1. Loading at source: Sucrose enters the sieve tubes by active transport. Companion cells supply the energy needed for this loading.

2. Entry of water: The high sucrose concentration in sieve tubes draws water from nearby xylem by osmosis. This greatly increases the fluid pressure inside the phloem at the source.

3. Bulk flow: The sucrose solution flows in bulk from the high-pressure source towards a lower-pressure sink. Gravity is not the driving force.

4. Unloading at sink: Sucrose is actively unloaded into sink tissues for respiration, growth or storage. Water also leaves the sieve tubes and returns towards the xylem by osmosis.

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5. Pressure difference continues: Unloading lowers pressure at the sink, while continued loading maintains high pressure at the source. This pressure difference keeps food moving from source to sink.

6. How do the plants excrete extra water and salts from their bodies?

Removal of extra water: During the day, plants lose excess water as vapours through stomatal, cuticular and lenticular transpiration. At night, when stomata are closed, many plants remove liquid water by **guttation**. Droplets of xylem sap are forced out through hydathodes at leaf tips or edges, usually by root pressure.

Removal or storage of extra salts: Some plants store excess salts or other mineral wastes as harmless crystals in cells, or in old leaves that later fall. In saline habitats, **halophytes** absorb salts to make their cell sap hypertonic, which prevents water from leaving their cells. They may store the extra salt in vacuoles or excrete it through special **salt glands** on their leaves.

These methods maintain a safe internal concentration of water and salts without a specialised excretory organ like an animal kidney.

7. Describe the process of gaseous exchange in plants

Gases move mainly by diffusion between the environment and plant cells. In leaves and young green stems, **stomata** allow air to enter and leave; inside the leaf, air spaces connect the internal cells. In woody stems, bark prevents direct exchange, so special pores called **lenticels** are used. Root and stem surfaces, and thin areas of epidermis and cuticle, also permit diffusion.

During daytime: All living cells respire, while green cells also photosynthesise. Photosynthesis uses the CO_2 produced by respiration and takes additional CO_2 from the environment. The O_2 produced by photosynthesis supplies respiration, and excess O_2 diffuses out through stomata. Thus, the net exchange in green parts is usually CO_2 in and O_2 out.

During night: Photosynthesis stops but respiration continues in every living cell. The plant therefore takes in O_2 from the environment and releases CO_2. Non-green organs such as roots respire all the time and must receive oxygen from air spaces in the soil.

8. Describe the mechanisms/adaptations in plants for excretion of wastes.

Plants produce fewer nitrogenous wastes than animals and use several simple methods to remove or isolate their metabolic wastes:

1. Gaseous wastes: Excess CO_2 from respiration and excess O_2 from photosynthesis diffuse out through general surfaces and stomata. Woody stems use lenticels.

2. Extra water: It is removed as vapour by transpiration or as liquid droplets by guttation.

3. Harmless crystals: Some plants convert wastes into insoluble, harmless crystals and store them in cells, so the wastes cannot interfere with metabolism.

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4. Storage in old leaves: Wastes may collect in leaves and are automatically removed when those leaves fall.

5. Excretion through special pores: Some wastes are forced out through bark or other pores. Rubber plants excrete **latex**, keekar (Acacia) excretes **gum**, coniferous trees excrete **resins**, and ladyfinger produces **mucilage**.

6. Salt removal: Halophytes may store salts in vacuoles or excrete them through salt glands on leaves. These adaptations keep the internal environment stable.

9. Explain osmotic adjustments in in hydrophytes, xerophytes and halophytes.

Hydrophytes: These plants live in freshwater or wet soil and can absorb water through their whole surface. Their problem is excess water, not shortage. Many have broad floating leaves with numerous stomata on the upper surface, through which water vapour escapes. They also remove extra water by guttation. Water lily is a common example.

Xerophytes: These plants live in very dry places. Deep roots absorb water from almost dry soil. A thick waxy cuticle and very few stomata reduce evaporation. Many xerophytes, such as cacti, store water in soft, juicy stems or roots called **succulent organs**. These features conserve water and help maintain cell turgor.

Halophytes: These plants live in sea water or salty marshes. The hypertonic environment tends to draw water out of their cells. They absorb salts and raise the solute concentration of their own cell sap, making it hypertonic relative to the surroundings, so water does not leave. Excess salt is stored safely in cells or excreted through salt glands on leaves. Many sea grasses show this adjustment.

D. Inquisitive Questions

1. Why do plants transpire more on a windy day compared to a humid one?

Wind removes the moist boundary layer of air around a leaf. The air outside therefore remains much drier than the air spaces inside the leaf, so water vapour diffuses rapidly through the stomata and transpiration increases.

On a humid day, the atmosphere already contains much water vapour. The concentration gradient between leaf air spaces and outside air becomes small, so diffusion slows. Therefore, moving dry air speeds transpiration, whereas humid air reduces it.

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E. More Important Questions (from the text)

1. Differentiate between active and passive transport in plants.

Active transport	Passive transport
Moves ions or molecules from lower to higher concentration.	Moves particles from higher to lower concentration, or water towards higher solute concentration.
Requires energy released by respiration.	Does not directly require metabolic energy.
Uses carrier proteins in the cell membrane.	Occurs by diffusion or osmosis without direct ATP use.
Example: uptake of mineral salts by root hairs and loading of sucrose into phloem.	Example: entry of water into root hairs by osmosis.

2. What is meant by source and sink in food translocation?

A **source** is a region that supplies food to the phloem. It may be a photosynthesising leaf or a storage organ releasing its reserves. A **sink** is a region that receives and uses or stores food, such as a growing root tip, young stem, developing fruit or storage root. Loading at the source and unloading at the sink create the pressure difference needed for translocation.

3. Why is transpiration called a necessary evil?

It is **evil** in the sense that a plant loses a large amount of absorbed water, and excessive loss may cause wilting. It is **necessary** because it creates the transpirational pull for water and mineral transport and cools the plant. Stomatal control and other adaptations reduce its harmful effects while retaining these benefits.

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Points to Remember

Mineral nutrients: Macronutrients are needed in large amounts; micronutrients in small amounts. Nitrogen is needed for proteins and chlorophyll, and magnesium is part of chlorophyll.

Root hairs: extensions of epidermal cells that increase absorption area. Mineral salts enter mainly by active transport and water by osmosis.

Root pathway: epidermis → cortex → endodermis → pericycle → xylem.

Xylem: carries water and salts mainly upwards. **Phloem:** translocates sucrose from source to sink.

Transpiration: loss of water vapour through stomata, cuticle or lenticels. Most occurs through stomata.

Stomatal opening: K^+ enters guard cells → water enters by osmosis → guard cells become turgid → pore opens. Reverse events close it.

Transpiration rate: rises with temperature and wind, but falls with high humidity.

Water ascent: transpirational pull, cohesion between water molecules and adhesion to xylem walls maintain the upward water column.

Pressure flow: active sucrose loading at source → osmotic water entry → high pressure → bulk flow to sink → unloading and pressure fall.

Gaseous exchange: stomata serve leaves and young stems; lenticels serve woody stems. At night plants take in O_2 and release CO_2.

Excretion: gases diffuse out; extra water leaves by transpiration and guttation; other wastes form crystals, collect in falling leaves, or leave as latex, gum, resin or mucilage.

Habitat adjustments: hydrophytes remove excess water; xerophytes conserve and store water; halophytes raise cell-sap solute concentration and remove excess salt.

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How to attempt this chapter in the paper

- Draw arrows for mechanisms: root uptake, stomatal movement, water ascent and pressure flow. An arrow sequence makes the answer clear and complete.
- Do not confuse osmosis with active transport: water enters root hairs by osmosis; mineral ions are taken up mainly by active transport.
- In source–sink answers, include active loading, osmosis, pressure difference, bulk flow and unloading.
- For every differentiation item, use a two-column table and give an example where possible.